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Optimal Monetary Policy in a Model with Agency Costs

This paper integrates a fully explicit model of agency costs into an otherwise standard Dynamic New Keynesian model in a particularly transparent way. A principal result is the characterization of agency costs as endogenous markup shocks in an output-gap version of the Phillips curve. The model's utility-based welfare criterion is derived explicitly and includes a measure of credit market tightness that we interpret as a risk premium. The paper also fully characterizes optimal monetary policy and provides conditions under which zero inflation is the optimal policy. Finally, optimal policy can be expressed as an inflation targeting criterion that (depending upon parameter values) can be either forward or backward looking.

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THE MACROECONOMIC EVENTS in the latter half of 2008 have sparked renewed interest in the role of financial shocks in the business cycle and the appropriate response of monetary policy to these shocks. This paper adds to this discussion by formally integrating a model of agency costs into an otherwise standard Dynamic New Keynesian (DNK) model. We do so in such a way that the agency–cost mechanism is quite transparent so that interactions between sticky prices and agency

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cost distortions are clearly identified. In addition our framework enables us to derive simple analytical expressions for the model-consistent welfare function.

We study the interaction of agency costs and sticky prices in a simple extension of the standard DNK model. Agency costs are modeled as a constraint on the firm's hiring of labor as in the holdup problem of Kiyotaki and Moore (1997). We assume that the entrepreneur's hiring of one productive factor (labor) is constrained by both entrepreneurial net worth and profits. More generally, the constraint proxies for the effect that asset prices and profits have on the ability of firms to finance operations. Net worth is accumulated over time via purchases of shares that are claims on the profit flow of sticky price firms that produce the final good. This leads to a natural interplay between price stickiness and collateral constraints. In our setup, monetary policy affects dividends and thus share prices by altering the profit flow of these sticky price firms. Share prices in turn affect the hiring of labor via the collateral constraint.

How should monetary policy be conducted in such an environment? From a public finance perspective, prices stickiness implies that real marginal cost acts like a distortionary *subsidy* on both factor inputs, while agency costs act like a distortionary *tax* on only one input (the constrained input). This suggests that a trade-off between stabilizing these two distortions may exist. We study this question formally by deriving the quadratic welfare function that is consistent with the underlying model and analyze optimal monetary policy in a linear-quadratic framework. Inflation and the output gap enter our loss function with the same coefficients as in the standard sticky price model. In addition, agency costs give rise to a new term that captures the variations in the tightness of the collateral constraint and that can more generally be interpreted as a risk premium. Thus, the recent concerns by central banks about credit market tightness and the volatility of risk premia have a counterpart in our welfare based loss function.

If credit constraints were absent from our model, we would obtain the standard result that central banks should fully stabilize inflation at all times in response to technology shocks. With agency costs and sticky prices, there is a special case where fully stabilizing inflation is once again optimal (all distortions are once again eliminated) when the economy is buffeted by only technology shocks. This special case requires preferences that are logarithmic in consumption and an efficient initial value for the model's state variable. In this case net worth happens to move by exactly the amount required to hire the first best quantity of labor when inflation is fully stabilized. Outside of this special case, it is not optimal to fully stabilize inflation in response to technology shocks. However, for technology shocks, the optimal deviation from inflation stability is small, so inflation stabilization is nearly optimal in this environment. Hence, the preferred interest rate rule should feature a strong anti-inflationary response, that is, an inflation coefficient that is well in excess of the value originally proposed by Taylor.

We model financial shocks as shocks to the net worth of entrepreneurs and show that they act like endogenous markup shocks. Net worth shocks imply that inflation stabilization comes at the cost of increased fluctuations in the output gap and in the tightness of the collateral constraint that enter our loss function. Consequently, a

temporary deviation from full price stability is warranted under optimal policy when financial sector shocks hit the economy. In addition because these financial shocks act as an endogenous markup we show that including the gap in a conventional Taylor rule is welfare reducing when the economy is buffeted by financial shocks.

We conclude this introduction with a brief review of the literature. Bernanke and Gertler (1989) provide the first attempt to build agency cost effects into an otherwise standard model of the business cycle. Their analysis is entirely qualitative. In a series of papers, Carlstrom and Fuerst (1997, 1998, 2001) build on this earlier work to make the analysis quantitative. These papers are in the tradition of flexible prices. Bernanke, Gertler, and Gilchrist (1999) take the next logical step to integrate these agency cost effects into a DNK model. These models help to quantify two distinct effects of agency costs: (i) a change in the economy's response to macroeconomic shocks because of the dynamics of borrower net worth and (ii) an additional source of shocks to the economy such as shocks to borrowers' balance sheets. But these previous analyses do not consider the question of optimal monetary policy.

Several recent papers analyze the performance of simple interest rate rules in agency cost models (e.g., Bernanke and Gertler 2001, Gilchrist and Leahy 2002, Faia and Monacelli 2007). Faia and Monacelli (2007) consider a model with sticky prices and credit frictions similar to Bernanke, Gertler, and Gilchrist (1999). They employ numerical second-order approximations to evaluate welfare of a parametric family of interest rate rules. Faia and Monacelli find that strict inflation targeting is the welfare maximizing policy rule within the restricted set of rules. They argue that the marginal benefit from neutralizing the price stickiness distortion largely outweighs the marginal benefit from neutralizing the credit friction distortion. Our linear-quadratic confirms this basic finding in the sense that the welfare weight on the risk premium is orders of magnitude smaller than the weight on inflation. In addition, Faia and Monacelli focus mainly on technology shocks. Our analysis considers shocks to net worth for which the optimal departure from price stability is larger than for productivity shocks.

Our work is most closely related to the papers by De Fiore and Tristani (2008) and Cúrdia and Woodford (2008). These authors also provide small-scale sticky price models with credit frictions and characterize optimal policy within a linear-quadratic framework. Cúrdia and Woodford focus on interest rate spreads between bank's lending and deposit rates that arise because loans are costly to produce. The analysis of these credit frictions is limited to a reduced-form relationship between credit spreads and macroeconomic conditions.

De Fiore and Tristani (2008) add a more complete underlying structure by including a costly state verification (CSV) framework within the standard DNK model. But De Fiore and Tristani abstract from the endogenous evolution of net worth by assuming that entrepreneurs receive a fixed endowment in every period. This assumption simplifies matters but eliminates the endogenous state variable that is of fundamental importance to the agency cost mechanism. Further, both of these previous papers do not feature any feedback between asset prices and net worth of credit-constrained

agents. In contrast, the interplay between the share price, net worth, and the agency cost distortion is central to the dynamics of our model.

The simplifying assumptions used in the paper make the analysis quite transparent but do have several undesirable features. First, the rigid collateral constraint implies that there are no defaults in equilibrium. Second, we can infer only a shadow value of the collateral constraint and not a true risk premium. Finally, the simplicity of the model leads to difficulties in calibration. These latter two features are partially remedied in the Appendix, which demonstrates a formal isomorphism between this model and a CSV framework. We use this link to calibrate the model to match a CSV framework.

This paper proceeds as follows. In the next section we outline the model, culminating in an expression for the welfare criterion. Section 3 analyzes the model quantitatively. Section 4 concludes.

1. THE MODEL

The model consists of households, entrepreneurs that produce intermediate goods, and sticky price firms that produce the final good. The basic structure is as follows. Households supply two types of labor input to the entrepreneurs. These entrepreneurs use a constant-returns-to-scale production function to produce an intermediate good using these two labor inputs. The entrepreneurial choice of one of the two labor inputs is constrained by accumulated net worth. This is the manifestation of agency costs in the model. The intermediate good is sold to monopolistically competitive final goods firms who use a linear production function to produce output. Pricing at this level is subject to Rotemberg (1982) adjustment costs. We now discuss each economic agent in turn.

1.1 Households

The typical household consumes the final good (c_t) and sells two types of labor input (L_t and u_t) to the entrepreneurs at factor prices w_t and r_t . Preferences are given by

$$U(C_t, L_t, u_t) = \frac{c_t^{1-\sigma}}{1-\sigma} - B_1 \frac{L_t^{1+\theta}}{1+\theta} - B_2 \frac{u_t^{1+\theta}}{1+\theta}.$$

This two-input framework can be interpreted in at least two ways. First, the aggregative household may be interpreted as a proxy for an economy with households that sell two distinct types of labor but that insure each other in terms of final consumption. This interprets u as a conventional unconstrained labor input and L as the labor input subject to credit constraints. Alternatively, the second labor input can be interpreted as “capital utilization.” Suppose that each household has a fixed level of capital, and each sells capital services to the entrepreneur at factor price r_t . Utilization

of the capital stock (u_t) causes a utility cost to the household. Both of these microlevel stories are consistent with the aggregative model analyzed here.

The household has access to two financial instruments. The first is a standard one-period bond. These are in zero net supply but are introduced to price the gross nominal interest rate denoted by R_t . Second, both households and entrepreneurs can purchase shares in the sticky price firm at price Q_t . These shares pay out dividends D_t . The aggregate supply of shares is normalized to unity. Ownership of these shares shifts back and forth endogenously between households and entrepreneurs. The input supplies, Fisher equation, and asset price equation are:

$$\frac{U_L(t)}{U_c(t)} = w_t(1 + w_{\text{sub}}) \quad (1)$$

$$\frac{U_u(t)}{U_c(t)} = r_t(1 + r_{\text{sub}}) \quad (2)$$

$$U_c(t) = \frac{E_t \beta U_c(t+1) R_t}{\pi_{t+1}} \quad (3)$$

$$Q_t U_c(t) = E_t \beta U_c(t+1) [Q_{t+1} + D_{t+1}], \quad (4)$$

where π_t denotes the gross inflation rate, w_{sub} and r_{sub} are steady-state wage subsidies, and E_t is the expectations operator.

1.2 Entrepreneurs

We assume that there are a continuum of long-lived entrepreneurs with linear consumption preferences and intertemporal discount rate β . Entrepreneurs hire two types of labor from households and use these inputs in a CRS production function to produce an intermediate good. Their profit function is given by

$$\text{profits}_t = p_t L_t^\alpha u_t^{1-\alpha} - w_t L_t - r_t u_t, \quad (5)$$

where p_t denotes the relative price of the intermediate good price (in terms of the final good), and $x_t \equiv L_t^\alpha u_t^{1-\alpha}$ denotes production of the intermediate good. The parameter, α , governs the importance of agency costs in the model. If $\alpha = 0$ the model collapses to the simple one-sector sticky price model without agency costs. As noted, this modeling formulation is isomorphic to a model with two sets of entrepreneurs, one using labor input L_t , and the other using labor input u_t . The former set of entrepreneurs is constrained by net worth, while the latter are not. These two inputs are then combined by a Cobb–Douglass production aggregator with coefficient α . In this case, the coefficient α can also be interpreted as the share of intermediate-good firms that are credit constrained.

Entrepreneurs face a collateral constraint on their hiring of the one labor input, L

$$w_t L_t \leq g(nw_t, p_t x_t - r_t u_t) \equiv n w_t^b (p_t x_t - r_t u_t)^{1-b}, \quad (6)$$

where $nw_t \equiv e_{t-1}(Q_t + D_t)$ denote the entrepreneur's net worth and e_{t-1} denotes the entrepreneur's equity share holdings. The collateral constraint (6) is motivated by a holdup problem as in Hart and Moore (1994). To fix intuition, consider the case of $b = 1$ in which only equity collateral can be used to satisfy the constraint. In this case the holdup problem would go something like this. After the L-input has been added to the production process, the entrepreneur must add his unique and inalienable human capital to the process for otherwise there would be no production. This means that *ex post* the entrepreneur can "hold up" the L-input supplier and force down her wages. To avoid this renegotiation problem, the L-input supplier requires the entrepreneur to back the promised wages with assets that can be seized if necessary.

We generalize this idea with a Cobb–Douglass constraint over collateral and operating profit. Now the L-input wage-bill is backed by a combination of existing collateral and the profit flow left over after paying the u-suppliers. Three comments are in order. First, the model will be calibrated so that the collateral constraint will always bind in equilibrium.¹ Second, one should interpret this constraint as holding in the neighborhood of the steady state in which net worth and profit flows are nonzero. Evidently the constraint needs modification for large deviations from the steady state. Third, the appendix demonstrates that a CSV framework implies a constraint identical (up to log deviations) to (6). The intuition for this result comes from a simple rewriting of (6) as

$$w_t L_t \leq l_t n w_t, \quad (7)$$

where $l_t \equiv \left(\frac{p_t x_t - r_t u_t}{w_t L_t} \right)^{\frac{1-b}{b}}$ is the leverage between net worth and loan size. The ratio in parentheses is the profit potential from the use of the L-input. Not surprisingly the equilibrium leverage is increasing in this ratio. Written this way, the constraint resembles what arises in a CSV framework. The Appendix demonstrates formally this isomorphism (up to log deviations and ignoring steady-state bankruptcy costs). Later we will use this isomorphism to aid in calibration.

Let ϕ_t denote the multiplier on the constraint (6). The optimization conditions include

$$\alpha p_t x_t = w_t L_t (1 + b\phi_t) \quad (8)$$

$$(1 - \alpha) p_t x_t = r_t u_t. \quad (9)$$

There is an alternative interpretation of $b\phi_t$ that now suggests itself. From (8), $b\phi_t$ is isomorphic to a model in which the L-input wage bill must be paid in advance of production at real interest rate cost $b\phi_t$. Compared to the unfettered input choice (9), $b\phi_t$ can thus be interpreted as a credit distortion on an intratemporal loan, and

1. More precisely, we log-linearize the model around a steady state in which this constraint binds and assume that it is also binding for small fluctuations around the steady state. Iacoviello (2005) examines the possibility that collateral constraints could be only occasionally binding and concludes that this is quite unlikely in his environment given a reasonable calibration of the volatility of the shocks.

fluctuations in $b\phi_t$ are thus fluctuations in the credit distortion.² Assuming that the collateral constraint binds, one can express (8) as

$$(1 + b\phi_t) = \left(\frac{\alpha p_t x_t}{n w_t} \right)^b,$$

so that the credit distortion is positively related to the output to net worth ratio. This relationship is similar to the reduced-form financial accelerator used in Gilchrist, Ortiz, and Zakrajšek (2009) (see their equation (24)). The key difference is what the entrepreneur borrows to finance: the entire capital stock in Gilchrist, Ortiz, and Zakrajšek, or one of the labor inputs as in this model. Therefore, b is a measure of the elasticity of the credit distortion to the project-size-to-net-worth ratio.

Because of the collateral constraint firms will earn profits in equilibrium. These profits are given by

$$profits_t = \alpha p_t x_t - w_t L_t = \alpha p_t x_t \left(\frac{b\phi_t}{1 + b\phi_t} \right). \quad (10)$$

In the neighborhood of the steady-state, profits are positive such that the rate of return on internal funds exceeds the time preference rate. Entrepreneur will thus desire to accumulate more net worth and postpone consumption. The entrepreneur's budget constraint is given by

$$c_t^e + e_t Q_t \leq e_{t-1} (Q_t + D_t) + profits_t. \quad (11)$$

Using the binding collateral constraint and equilibrium profits we have

$$c_t^e + e_t Q_t \leq \alpha p_t x_t F(\phi_t), \quad (12)$$

where

$$F(\phi_t) \equiv \left[\left(\frac{b\phi_t}{1 + b\phi_t} \right) + \left(\frac{1}{1 + b\phi_t} \right)^{\frac{1}{b}} \right].$$

Linear preferences along with the profit potential from acquiring greater net worth to mitigate future agency costs implies that entrepreneurial consumption will always be zero in equilibrium. To ensure that entrepreneurs do not acquire so much collateral that the collateral constraint does not bind, we assume that fraction $(1 - \gamma)$ of the entrepreneurs die each period, to be replaced by an equal number of new entrepreneurs each endowed with a trivial amount of net worth. These deaths are unexpected so the

2. The implicit interest cost of utilizing the L-input is similar to the cost channel of monetary transmission (see Barth and Ramey 2001, Ravenna and Walsh 2006). In these earlier papers the firm's wage bill is subject to a cash-in-advance constraint so that marginal production costs are grossed up by the nominal interest rate. In the present paper there is a "collateral-in-advance" constraint (6). The implicit interest cost is a real rate, and the severity of the constraint is endogenous via entrepreneurial wealth accumulation.

entrepreneurs do not consume their assets before dying. After dying, all assets of the entrepreneurs are redistributed to households.³ This simplifying assumption implies that the model's quadratic welfare function extends the standard loss function with only one additional term that accounts for agency costs.⁴

1.3 Sticky Price Firms

Monopolistically competitive firms indexed by j produce final goods $y_{t,j}$ that are aggregated to an output bundle according to $y_t = [\int_0^1 y_{t,j}^{(\varepsilon-1)/\varepsilon} dj]^{\varepsilon/(\varepsilon-1)}$ with $\varepsilon > 1$. The final goods firms purchase the intermediate good from entrepreneurs at relative price p_t . The production function is given by $y_{t,j} = a_t x_{t,j}$, where productivity a_t follows an exogenous AR(1) process. Since the production function is linear, real marginal cost is given by $z_t = \frac{p_t}{a_t}$. The final goods price is subject to a Rotemberg-style (1982) quadratic cost of price adjustment, which enter the profit function of firm j as $\frac{\varphi}{2} [\frac{p_{t,j} - p_{t-1,j}}{p_{t-1,j}}]^2 y_t$, with $\varphi \geq 0$. These costs disappear from the linearized version of the social resource constraint as long as the gross inflation rate is unity in the steady state. Proposition 1 demonstrates that it is indeed optimal for a central bank that adopts a present value welfare criterion to deliver such a steady-state inflation rate.

We use hatted variables throughout to denote their percentage deviation from the deterministic steady state. In a symmetric equilibrium, the Rotemberg price-setting problem gives rise to the standard DNK Phillips curve given by

$$\hat{\pi}_t = \lambda \hat{z}_t + \beta E_t \hat{\pi}_{t+1} + \lambda \epsilon_t^\pi, \quad (13)$$

where $\lambda \equiv (\varepsilon - 1)/\varphi$ and ϵ_t^π is a mark-up shock with standard deviation σ_π . Monopolistic competition implies that these firms earn profits in equilibrium. These profits are paid out as dividends to shareholders of the sticky price firms. These dividends are given by $D_t = a_t x_t (1 - z_t)$.

1.4 A Tale of Two Distortions

A helpful interpretation of the model is revealed by considering the two labor input choices.

$$\frac{U_L(t)}{U_C(t)} = \frac{z_t}{(1 + b\phi_t)} MPL_t (1 + w_{\text{sub}}) \quad (14)$$

3. See Carlstrom and Fuerst (1998) for a discussion of other methods of ensuring that the collateral constraint binds in equilibrium. For example, as in Carlstrom and Fuerst (1997), we could assume that entrepreneurs are risk neutral and choose consumption optimally. To ensure that the collateral constraint binds, we could assume that entrepreneurs discount the future more heavily than do households. The Euler equation for entrepreneurial intertemporal consumption would be given by: $Q_t = \beta \gamma E_t (Q_{t+1} + D_{t+1})(1 + b\phi_{t+1})$, where $\gamma < 1$ is the higher rate of discount. In log-deviations we would have: $\hat{q}_t = \beta E_t \hat{q}_{t+1} + (1 - \beta) E_t \hat{d}_{t+1} + E_t b \hat{\phi}_{t+1}$. We do not follow this approach as we would then need to include entrepreneurial consumption in the model, which would needlessly complicate the welfare analysis.

4. This is in contrast to De Fiore and Tristani (2008) who include entrepreneurial consumption directly in their welfare analysis.

$$\frac{U_u(t)}{U_c(t)} = z_t MPU_t (1 + r_{\text{sub}}). \quad (15)$$

We introduce wage subsidies to render the steady-state of the model efficient. The Appendix provides the details. Here, MPU and MPL denote the marginal product of u and L , respectively, while w_{sub} and r_{sub} are the steady state subsidies. Increases in real marginal cost (z_t) are isomorphic to a decrease in a distortionary tax on all labor income, while an increase in the agency cost distortion ($b\phi_t$) is isomorphic to an increase in a distortionary tax on labor income only from the choice of constrained labor, L_t . Hence, increases in z_t will increase output via *symmetric* movements in both types of labor, while increases in $b\phi_t$ will decrease output via an *asymmetric* decline in L_t .

This tale of two distortions has an immediate implication in terms of optimal (fiscal) policy. The efficient allocations can be obtained with a variable subsidy policy such that

$$\frac{z_t}{(1 + b\phi_t)}(1 + w_{\text{sub},t}) = 1$$

$$z_t(1 + r_{\text{sub},t}) = 1.$$

We simply note this in passing and do not pursue it further because: (i) such a finely tuned fiscal policy strains credulity and (ii) our interest is in monetary policy.

The Appendix outlines the entire model in log deviations. For present purposes, we collapse the model into a smaller system that has a clear economic interpretation in terms of these two distortions. Let us begin by defining the output gap, etc., to be the difference in log deviations between actual output and the level of output when there are no price or credit frictions, $\hat{y}_t^g \equiv (\hat{y}_t - \hat{y}_t^{\text{eff}})$, where

$$\hat{y}_t^{\text{eff}} = \frac{(1 + \theta)}{\sigma + \theta} a_t. \quad (16)$$

In this case, (A1)–(A2), (A7)–(A8), and (A11) can be combined to yield

$$\hat{u}_t^g = \frac{1}{\sigma + \theta} \hat{z}_t + \frac{\alpha(\sigma - 1)}{(\sigma + \theta)(1 + \theta)} b\hat{\phi}_t \quad (17)$$

$$\hat{L}_t^g = \frac{1}{\sigma + \theta} \hat{z}_t - \left[\frac{\sigma(1 - \alpha) + (\alpha + \theta)}{(\sigma + \theta)(1 + \theta)} \right] b\hat{\phi}_t \quad (18)$$

$$\hat{y}_t^g = \frac{1}{\sigma + \theta} \hat{z}_t - \frac{\alpha}{(\sigma + \theta)} b\hat{\phi}_t \quad (19)$$

$$\hat{w}_t^g = \sigma \hat{y}_t^g + \theta \hat{L}_t^g \quad (20)$$

$$\hat{r}_t^g = \sigma \hat{y}_t^g + \theta \hat{u}_t^g. \quad (21)$$

As noted earlier, output varies positively with marginal cost, and negatively with the agency cost distortion. Marginal cost affects both inputs symmetrically, and increases in the agency cost distortion necessarily lowers L_t while having an ambiguous effect on u_t (depending upon the size of σ). The Phillips curve (A9) and Fisher equation (A3) are given by

$$\hat{\pi}_t = \lambda \hat{z}_t + \beta E_t \hat{\pi}_{t+1} + \lambda \epsilon_t^\pi \quad (22)$$

$$\frac{\sigma}{\sigma + \theta} (E_t \hat{z}_{t+1} - \hat{z}_t) - \frac{\alpha \sigma}{\sigma + \theta} (E_t b \hat{\phi}_{t+1} - b \hat{\phi}_t) = \hat{R}_t - E_t \hat{\pi}_{t+1} - \hat{\psi}_t, \quad (23)$$

where $(\hat{\psi}_t \equiv \sigma(E_t \hat{y}_{t+1}^{\text{eff}} - \hat{y}_t^{\text{eff}}))$ is an exogenous fluctuation in the natural rate of interest. The agency cost movements $b \hat{\phi}_t$ are endogenous to the model, so to close the model we need an expression for the dynamics of the agency cost distortion. As a first step, we use (A5) to solve for the asset price

$$\hat{q}_t = \frac{1}{(\sigma + \theta)} [(1 + \sigma + \theta) \hat{z}_t - [\alpha b - \Lambda(\sigma + \theta)] \hat{\phi}_t + (1 + \theta) \hat{a}_t] - \hat{e}_t, \quad (24)$$

where $\Lambda \equiv \frac{F'}{F} \approx (b - 1) \leq 0$. Equations (A6) and (A10) then provide a dynamic equation in $\hat{e}_t$

$$\hat{e}_t = \frac{1}{\beta} [\hat{e}_{t-1} + \hat{n}_t - \varepsilon(1 - \beta) \hat{z}_t + (1 + \beta \Lambda) \hat{\phi}_t], \quad (25)$$

where $\hat{n}_t$ is an exogenous shock to the net worth of entrepreneurs with variance σ_n^2 . Since $\hat{e}_{t-1}$ is the fraction of shares owned by entrepreneurs, $\hat{n}_t$ is a redistribution of wealth from households to entrepreneurs. The last dynamic relationship comes from using the asset-pricing formula (A4) and equation (25) scrolled forward

$$\begin{aligned} [\alpha b(1 - \sigma) - \Lambda(\sigma + \theta)] \hat{\phi}_t &= [\alpha b + \theta + \sigma(1 - \alpha b)] E_t \phi_{t+1} - (1 + \theta) E_t \Delta \hat{z}_{t+1} \\ &\quad + (\sigma - 1)(1 + \theta) E_t \Delta \hat{a}_{t+1} + (\sigma + \theta) E_t \hat{n}_{t+1}. \end{aligned} \quad (26)$$

This completes the model.

In summary, the model is given by the two familiar DNK equations (22)–(23), and equations (25) and (26) that track the dynamic behavior of entrepreneurial wealth and the agency cost distortion. The model is then closed by articulating a rule for monetary policy. Several comments are in order.

First, if we set $\alpha = 0$ so that the constrained input is not part of the production process, the model collapses to the familiar DNK model without the agency cost distortion.⁵ Hence, it is quite easy to consider the model with and without this distortion.

5. This statement is true assuming that the central bank interest rate rule does not include a response to endogenous variables in (24)–(26) such as share prices or the risk premium.

Second, agency costs manifest themselves as a distortion, $b\hat{\phi}_t$, in the Fisher equation (23). Fluctuations in $b\hat{\phi}_t$ lead to an inefficient input mix (17)–(18), and thus fluctuations in the output gap (19). As noted earlier, one can interpret these movements as fluctuations in the credit distortion on loans to finance the L-input. These effects are scaled by the production coefficient α .

Finally, the addition of agency costs has no effect on the standard DNK pricing relationship (22). Inflation is still driven by fluctuations in marginal cost. However, if we write the pricing relationship in terms of the output gap (using (19)) we have

$$\hat{\pi}_t = \lambda(\sigma + \theta)\hat{y}_t^g + \alpha\lambda b\hat{\phi}_t + \beta E_t\hat{\pi}_{t+1} + \lambda\epsilon_t^\pi. \quad (29)$$

Fluctuations in agency costs affect inflation in a manner symmetric to markup shocks, but with the key difference that agency costs evolve endogenously and are not simply exogenous shocks tacked on to the pricing relationship. Hence, whether agency costs need to be considered when estimating a Phillips curve depends upon whether marginal cost or the output gap are part of the empirical estimation. Relatedly, even if there are no markup shocks (ϵ_t^π), a monetary policy that stabilizes the output gap does not stabilize inflation.

1.5 Endogenous Dynamics with Flexible Prices

In the standard DNK model there are no endogenous state variables but agency costs naturally introduce net worth as an endogenous state variable. Hence, to understand the agency cost mechanism one needs to examine the endogenous dynamics of entrepreneurial net worth—to what extent is current net worth propagated forward? The dynamic mechanism is particularly transparent for the case of flexible prices. Note that in this DNK framework, these flexible price dynamics are identical to a sticky price model in which the central bank stabilizes the inflation rate (assuming that there are no markup shocks). As we will see later, a stable inflation rate is nearly optimal so that this special case is particularly interesting.

With stable prices (and no exogenous markup shocks), equations (25) and (26) become

$$(1 + \beta\Lambda)\hat{\phi}_t = \beta\hat{e}_t - (\hat{e}_{t-1} + \hat{n}_t) \quad (30)$$

$$E_t\hat{\phi}_{t+1} = A\hat{\phi}_t + \frac{(\sigma - 1)(1 + \theta)(1 - \rho)}{\alpha b + \theta + \sigma(1 - \alpha b)}\hat{a}_t - \frac{\rho_n(\sigma + \theta)}{\alpha b + \theta + \sigma(1 - \alpha b)}\hat{n}_t, \quad (31)$$

where $A \equiv \frac{\alpha b(1 - \sigma) - \Lambda(\sigma + \theta)}{\alpha b(1 - \sigma) + (\sigma + \theta)} \leq 1$, $\Lambda \equiv \frac{F'}{F} \approx (b - 1) \leq 0$, and (ρ, ρ_n) are the autocorrelation in the productivity and net worth shock, respectively. The endogenous agency cost dynamics are given by the system's stable eigenvalue, $A < 1$ (the other root is $1/\beta$). (The eigenvalue A is also the response of $\hat{e}_t$ to $\hat{e}_{t-1}$.) A stationary equilibrium exists if and only if $A > -1$, or $2\alpha b(\sigma - 1) < (1 - \Lambda)(\sigma + \theta)$. The endogenous agency cost dynamics are nonoscillatory ($A > 0$) if σ is not too large, that is, if

$\sigma(\alpha b + \Lambda) < \alpha b - \Lambda\theta$. If the opposite inequality holds, then there are damped oscillations. Similarly, as long as b is not too close to 1, oscillations will not occur.⁶ As b gets very small A approaches 1.

In this flexible price environment, the behavior of output and the gap depend solely on the behavior of the credit distortion (see (16) and (19)). The response of the credit distortion $b\hat{\phi}_t$ to shocks is given by the equilibrium decision rule

$$\begin{aligned} \hat{\phi}_t = & \frac{\beta A - 1}{1 + \beta \Lambda} e_{t-1} + \left[\frac{\beta A - 1}{(1 + \beta \Lambda)} + \frac{\beta p_n (\sigma + \theta)}{(\alpha b + \theta + \sigma (1 - \alpha b))} \right] \frac{n_t}{(1 - \beta \rho_n)} \\ & + \frac{\beta (1 - \sigma) (1 + \theta) (1 - \rho_a)}{(1 - \beta \rho_a) (\alpha b + \theta + \sigma (1 - \alpha b))} a_t. \end{aligned}$$

The first term in braces is negative so that (not surprisingly) the credit distortion declines (and output increases) with increases in net worth as these shocks relax the collateral constraint. Note that when $\rho_n = 0$, the response to the net worth shock is identical to the response to lagged entrepreneurial share holdings. But the response to productivity shocks depends on the size of σ . There are two effects here. First, an increase in productivity increases share prices thus relaxing the constraint. Second, the higher productivity increases real wages and the entrepreneur's L-input wage bill, thus tightening the constraint. If $\sigma = 1$, these two effects cancel. If $\sigma < 1$, the wage bill moves by more than the share price so that the credit distortion rises with positive technology shocks.⁷ The opposite is true, however, for $\sigma > 1$.

To summarize, the response of the credit distortions to net worth shocks is countercyclical, while the response to productivity shocks can be either pro- or countercyclical depending upon the size of σ . Of course, the unconditional cyclicity of the risk premium depends on the relative importance of the different types of shocks.

Before proceeding to the welfare analysis, (30)–(31) demonstrate a benchmark result. Recall that (absent markup shocks) these equations also describe the model under the case of sticky prices if the central bank stabilizes the inflation rate. Stabilizing the inflation rate also stabilizes marginal cost (22). For the case of $\sigma = 1$, equations (30)–(31) imply that the credit distortion does not respond to technology shocks. Hence, for the case of technology shocks and $\sigma = 1$, if the economy begins at the steady state ($\hat{e}_{t-1} = 0$), then a stabilizing inflation eliminates both the marginal cost and agency cost distortion and is thus the optimal policy.

6. In an earlier version of the paper we assumed that $b = 1$ ($\Lambda = 0$) so that entrepreneurial borrowing is only constrained by net worth. In this case there are oscillations if and only if $\sigma > 1$. But it does not take much movement from this strict ($b = 1$) collateral constraint to eliminate oscillations. For example, if $\sigma = \theta = 2$ and $\alpha = 1/2$, as long as $b < 0.89$ there will not be oscillatory dynamics. The presence of sticky prices has only modest effects on this cutoff.

7. This is the same result that Carlstrom and Fuerst (1997, 1998) obtain in their model with physical capital and investment spending. Evidently, a model with investment spending and $\sigma > 1$ behaves like a model without investment spending but $\sigma < 1$. This suggests that the results of Carlstrom and Fuerst (1997, 1998) on the behavior of the risk premium are sensitive to the inclusion of capital adjustment costs.

1.6 Welfare Criterion

The transparency of the model allows for analytical expressions for the loss function. We first note that zero is the optimal steady-state inflation rate. The subsequent proposition provides two analytical expressions for the welfare function. The first is expressed in terms of the output gap, the second in the underlying distortions.

PROPOSITION 1. *The gross inflation rate that maximizes a present value welfare criterion is unity in the steady state regardless of whether subsidies exist that make the steady state efficient or not.*

PROOF. See the Appendix.

PROPOSITION 2. *A quadratic approximation to the welfare function around a steady state with zero inflation is given by $\frac{U(t)-U^*(t)}{U_c(ss)c_{ss}} \approx -\frac{1}{2} \left[\frac{(\varepsilon-1)}{\lambda} \hat{\pi}_t^2 + (\sigma + \theta)(\hat{y}_t^g)^2 + \frac{\alpha(1-\alpha)}{1+\theta} (b\hat{\phi}_t)^2 \right]$, where $U^*(t)$ denotes welfare in a first best economy without agency or price adjustment costs. Using (17), we can alternatively express this in terms of the two underlying distortions*

$$\begin{aligned} \frac{U(t) - U^*(t)}{U_c(ss)c_{ss}} \approx & -\frac{1}{2} \left[\frac{(\varepsilon - 1)}{\lambda} \hat{\pi}_t^2 + \frac{\alpha [\theta + \alpha + \sigma (1 - \alpha)]}{(1 + \theta)(\sigma + \theta)} (b\hat{\phi}_t)^2 \right. \\ & \left. + \frac{1}{(\sigma + \theta)} \hat{z}_t^2 - \frac{2\alpha}{(\sigma + \theta)} b\hat{\phi}_t \hat{z}_t \right]. \end{aligned}$$

PROOF. See the Appendix.

Proposition 2 has several important implications. First, agency costs add the variance of $b\hat{\phi}_t$ to the loss function, which captures the tightness of the collateral constraint. We can loosely think of $b\hat{\phi}_t$ as related to the risk premia so that traditional concerns by central banks about credit market tightness and the volatility of risk premia have a counterpart in our welfare-based loss function. When $\alpha = 0$, the constrained labor input is not needed for production and the collateral constraint is not operative. Hence, our loss function collapses to the standard loss function as in Woodford (2003).⁸ For $\alpha > 0$, the gap is no longer a sufficient statistic for welfare purposes as the composition of output (u-produced versus L-produced) is relevant. When $\alpha = 1$, all output is L-produced, the credit distortion drops out of the gap version of the welfare criterion, and the gap (along with inflation) becomes a sufficient statistic for welfare since the relative distortion between constrained and unconstrained labor is no longer operative.

8. The standard DNK welfare criterion assumes subsidies are paid to the firm to make marginal cost equal to unity in the steady state. This implies that the welfare weight on inflation is $\frac{\varepsilon}{\lambda}$. The weight here differs slightly because marginal cost is less than unity in the steady state. Instead, the subsidies we introduce to render the steady-state efficient are paid directly to households to boost labor supply. See Keen and Wang (2004) for details of the Rotemberg approximation.

Second, writing the objective function in terms of marginal cost implies a preference for the two underlying distortions, $b\hat{\phi}_t$ and $\hat{z}_t$, to positively covary. This is not surprising as $b\hat{\phi}_t$ acts as a tax and $\hat{z}_t$ acts as a subsidy to labor.

Finally, the importance of stabilizing inflation is quite clear. For typical calibrations, the coefficient on inflation is at least an order of magnitude larger than the coefficients on the credit distortion and output gap. For example, for our baseline calibration, the weight on inflation is roughly 1,000 times larger than the weight on the credit distortion. Hence, the loss function implies a strong preference for stabilizing inflation, even at the cost of allowing the other distortions to vary. This is the principal reason that zero inflation is close to the optimal policy in a quantitative sense. For this reason we consider an alternative experiment later where ε is very small so that the weight on inflation is much smaller (we also change φ to keep the slope of the Phillips curve the same).

2. OPTIMAL POLICY AND SIMPLE RULES

In this section, we provide some analytical and quantitative results on optimal policy and the welfare losses of alternative policy rules. Under commitment, the policy maker maximizes

$$\max_{\{\hat{\pi}_t, \hat{\phi}_t, \hat{y}_t^g, \hat{e}_t\}_{t=0}^{\infty}} -\frac{1}{2}E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(\varepsilon - 1)}{\lambda} \hat{\pi}_t^2 + (\sigma + \theta) (\hat{y}_t^g)^2 + D \frac{\alpha(1 - \alpha)}{1 + \theta} (b\hat{\phi}_t)^2 \right], \quad (32)$$

subject to the constraints

$$\hat{\pi}_t = \lambda(\sigma + \theta) \hat{y}_t^g + \alpha \lambda b \hat{\phi}_t + \beta E_t \hat{\pi}_{t+1} + \lambda \epsilon_t^\pi, \quad (33)$$

$$\hat{e}_t = \frac{1}{\beta} \left\{ \hat{e}_{t-1} - \varepsilon(1 - \beta)(\sigma + \theta) \hat{y}_t^g + [1 + \beta\Lambda - \alpha\varepsilon(1 - \beta)b] \hat{\phi}_t + n_t \right\}, \quad (34)$$

$$\begin{aligned} \hat{\phi}_t &= \frac{(1 + \theta)}{(1 + \Lambda)} E_t \Delta \hat{y}_{t+1}^g - \frac{(1 - \alpha b)}{(1 + \Lambda)} E_t \Delta \hat{\phi}_{t+1} \\ &\quad - \frac{(\sigma - 1)(1 + \theta)}{(\sigma + \theta)(1 + \Lambda)} E_t \Delta \hat{a}_{t+1} - \frac{1}{(1 + \Lambda)} E_t n_{t+1}, \end{aligned} \quad (35)$$

where (34)–(35) are (25)–(26) written in terms of the output gap. We assume that the policymaker is constrained by commitments in the initial period. See Woodford (2003) for details on the timeless perspective.

Note that we have introduced the constant D into the objective (32). For $D = 1$, the objective is identical to the true welfare criterion in the model. For $D = 0$, the policymaker ignores the welfare consequences of the credit distortion although he does not ignore the credit distortion in the constraints (33)–(35). This device allows

us to distinguish between changes in optimal policy that result from the change in the objective function versus changes in the structural model.

PROPOSITION 3. *The optimal inflation target under commitment from a timeless perspective is given by*

$$\begin{aligned} \hat{\pi}_t = & \left[\frac{\alpha b \varepsilon (1 - \beta) - (1 + \beta \Lambda)}{(\varepsilon - 1)(1 + \beta \Lambda)} \right] \Delta \hat{y}_t^g - D \left[\frac{\alpha b \varepsilon (1 - \beta)(1 - \alpha)}{(\varepsilon - 1)(1 + \theta)(1 + \beta \Lambda)} \right] b \Delta \hat{\phi}_t \\ & - \left[\frac{\varepsilon(1 - \beta)[\alpha b(\sigma - 1) + (\sigma + \theta)\Lambda] + (1 + \theta)(1 + \beta \Lambda)}{(\varepsilon - 1)(\sigma + \theta)(1 + \beta \Lambda)(1 + \Lambda)} \right] \hat{X}_t \\ & + \left[\frac{\varepsilon(1 - \beta)[\alpha b(\sigma - 1) - (\sigma + \theta)] + (1 + \theta)(1 + \beta \Lambda)}{\beta(\varepsilon - 1)(\sigma + \theta)(1 + \beta \Lambda)(1 + \Lambda)} \right] \hat{X}_{t-1} \end{aligned}$$

The variable $\hat{X}_t$ is a lag-polynomial in $\Delta \hat{\phi}_t$ and $\Delta \hat{y}_t^g$ defined via the recursion:

$$\begin{aligned} \hat{X}_t &= \beta A E_t \hat{X}_{t+1} + E_t b \Xi_{t+1} \\ \Xi_t &\equiv \frac{\alpha \beta (\sigma + \theta)(1 + \Lambda) [(1 + \theta) \Delta y_t^g - D(1 - \alpha) b \Delta \hat{\phi}_t]}{(1 + \theta)(\sigma + \theta + \alpha b(1 - \sigma))}. \end{aligned}$$

Recall that $A \equiv \frac{\alpha b(1 - \sigma) - \Lambda(\sigma + \theta)}{\alpha b(1 - \sigma) + (\sigma + \theta)} < 1$, $\Lambda = \frac{F'}{F} \approx (b - 1) \leq 0$.

PROOF. See the Appendix.

Several comments are in order. First, optimal policy relates inflation to the *first difference* of the remaining variables in the loss function rather than to the levels. This feature is shared with the targeting criterion in the simple DNK model. Note that for $\alpha = 0$, this condition is simply the targeting criterion from the standard DNK model, $\pi_t = -\frac{1}{\varepsilon - 1} \Delta y_t^{\text{gap}}$. At the other extreme, if $\alpha = 1$, then the credit distortion drops out of the targeting criterion because (as noted earlier) the output gap is a sufficient statistic for welfare purposes. But the targeting criterion still differs from the standard DNK model because of the endogenous behavior of net worth.

Second, the recursion in $\hat{X}_t$ implies that policy will be either forward looking or backward looking depending on the size of βA . If $\beta A < -1$, policy is backward looking with

$$\hat{X}_t = -b \sum_{j=0}^{\infty} [(\beta A)^{-1}]^{j+1} \Xi_{t-j}.$$

The more likely case is that βA is in the unit circle ($\beta A > -1$) so that policy is forward looking with

$$\hat{X}_t = b \sum_{j=0}^{\infty} (\beta A)^j E_t \Xi_{t+1+j}.$$

The weight on the $t+1$ forecast is b . The sum of all future coefficients beyond one period ahead is given by

$$b \sum_{j=1}^{\infty} (\beta A)^j = \frac{b\beta [\alpha b(1-\sigma) - \Lambda(\sigma + \theta)]}{(1 + \beta\Lambda)(\sigma + \theta) + (1 - \beta)\alpha b(1 - \sigma)} \approx \frac{\alpha b(1 - \sigma)}{(\sigma + \theta)} + (1 - b).$$

For $\sigma = 1$, the weight placed on forecasts beyond one period ahead is $1 - b$. Hence, for a strict Kiyotaki and Moore constraint, the degree of forward-looking behavior (beyond one period) is modest and directly proportional to the degree of agency costs (the size of α).

Finally, for $\sigma = b = \beta = 1$ the targeting criterion is particularly simple in that only one lead and lag is included in the expression. We have

$$\begin{aligned} \hat{\pi}_t = & -\frac{1}{(\varepsilon - 1)} \Delta \hat{y}_t^g - \frac{\alpha}{(\varepsilon - 1)} \{E_t \Delta \hat{y}_{t+1}^g - E_{t-1} \Delta \hat{y}_t^g\} \\ & + \frac{D\alpha(1 - \alpha)}{(\varepsilon - 1)(1 + \theta)} \{E_t \Delta \hat{\phi}_{t+1} - E_{t-1} \Delta \hat{\phi}_t\}. \end{aligned}$$

In comparison to the standard DNK criterion ($\hat{\pi}_t = -\frac{1}{(\varepsilon-1)} \Delta \hat{y}_t^g$), the optimal inflation target now responds to future and lagged gap growth. This is because of the presence of an endogenous state variable in the credit model and occurs even when the central banker's objective does not include the credit distortion ($D = 0$). Similarly if $\alpha = D = 1$, then policy does not respond to the credit distortion directly as gap growth becomes a sufficient statistic for welfare purposes. For $D = 1$ and when α is between zero and one, optimal policy also responds to changes in the credit distortion.

2.1 Calibration

We now turn to a quantitative analysis of optimal policy. Parameters from the familiar DNK model are set to standard values. The discount factor β is set 0.99. We set the elasticity in the CES aggregator $\varepsilon = 10$, consistent with a markup of roughly 11%. We calibrate the Rotemberg price adjustment cost parameter such that the coefficient on marginal cost in the Phillip curve is 0.052. This resulting value for φ is 173.08. In a log-linearly equivalent Calvo price setting model, such a slope implies that prices are fixed for five quarters on average.

Our calibration for the preference parameters follows Woodford (2003, p. 172): $\sigma = 0.16$ and $\theta = 0.47$, implying large intertemporal and intratemporal elasticities. This implies that there are strategic complementarities in the model. Woodford argues that in a model without physical capital and investment spending, one needs a high intertemporal elasticity to match the interest sensitivity of GDP. We assume that the total factor productivity follows an AR(1) process with autoregressive coefficient 0.95, whereas net worth and markup processes are assumed to be slightly less persistent with AR coefficients 0.9.

The credit-related parameters are α , b , and the steady-state value of the collateral constraint ϕ_{ss} . As noted earlier, α can be interpreted as the share of intermediate goods firms that are collateral constrained. We think of these as being “small” firms.

Impulse response to monetary shock

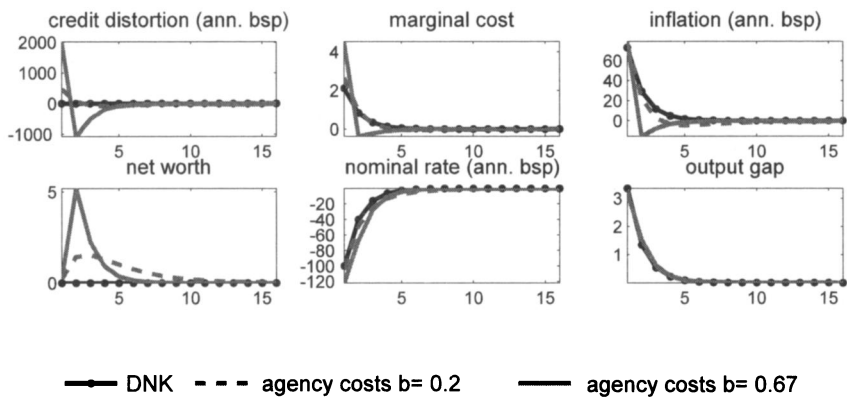


FIG. 1. Impulse Response to a Monetary Shock.

NOTE: The x-axis measure quarters after the shock. The y-axis measures percentage deviations from the steady state for all variables except for the credit distortion, inflation, and the nominal rate where it measures the deviations in annualized basis points.

Using an employment level of 500 employees as a crude cutoff, we set $\alpha = 1/2$ as this matches the share of private sector employment in firms with less than 500 employees.

The Appendix demonstrates a formal linkage between this model and a CSV model. (Except for small bankruptcy costs in the resource constraint, the models are isomorphic.) We use this isomorphism to calibrate b and ϕ_{ss} . We calibrate the CSV model to match the risk premium for the constrained firms to the average spread between the BB+ and 10-year Treasury bond. From 1996 to the present, this is a spread of 330 annual basis points (bps). Since $(1 - \alpha) = 1/2$ of the firms in the model do not face any credit constraints, matching the risk premium to the BB+ rate seems reasonable. If you averaged this 330 bp spread with the zero spread for the nonconstrained firms, then the aggregate risk premium would be 165 bp, which is midway between the AAA risk spread and the prime rate risk spread. The CSV model (which we calibrate in the Appendix) implies $b = 0.20$, and $b\phi_{ss} = 0.026$. The value of the entrepreneurial death rate in the model is then chosen as a residual to hit the level of ϕ_{ss} . (See the Appendix for details.) As a form of sensitivity analysis, we also consider the case of $b = 2/3$. This is more in line with the Kiyotaki and Moore (1997) rigid collateral constraint, yet b is still small enough to ensure nonoscillatory dynamics.

2.2 Impulse Response Functions

We begin our quantitative investigation of the model in Figure 1 by considering the effect of a monetary shock. The monetary policy rule is assumed to be a Taylor interest rate rule given by $\hat{R}_t = \tau \hat{\pi}_t + \tau_g \hat{y}_t^g + \varepsilon_t^R$ (with $\tau = 1.5$, $\tau_g = 0.5$), where ε_t^R is the exogenous monetary policy shock with autocorrelation set to 0.4. Figure 1

presents the impulse response function to a decline in the nominal rate for the DNK model ($\alpha = 0$), and the two calibrations of the credit constraint model with $\alpha = 0.5$ and with $b = 0.2$ or $b = 0.67$. The size of the shock is chosen to cause the interest rate in the DNK model to fall by 100 annual basis points. This same shock size is used for the credit models, but the implied behavior of the nominal rate is slightly different because of the endogenous component of the Taylor rule.

Most interestingly, the behavior of the output gap is nearly identical for all three models. But the models do differ in the manner in which this gap behavior arises. In the two credit models the monetary expansion leads to: (i) an increased demand for labor and (ii) a decline in firm profits and share prices.⁹ Taken together the monetary shock thus exacerbates the credit distortion. This implies a sharper increase in marginal cost in these two models (compared to the DNK model). The endogenous wealth accumulation then leads to a sharp decline in the credit distortion so that the distortion and marginal cost fall sharply in the subsequent period. This then leads to a very different inflation behavior in the credit models relative to the DNK model. However, for our preferred calibration of $b = 0.2$, the inflation behavior is very similar to the DNK model.

What do these results suggest for monetary policy? The credit frictions do not significantly alter the effect of a monetary expansion on the output gap. However, these movements are more costly in a model with credit frictions as they lead to an inefficient production mix because of the endogenous credit distortion. For a central bank that cares about this distortion (as in the welfare function derived earlier), these results suggest that there will be more reticence to take policy actions to mitigate gap movements.

Figures 2(a)–(b) present the impulse responses to a negative technology shock for the two credit market calibrations under three different versions of monetary policy: a Taylor interest rate rule given by $\hat{R}_t = \tau \hat{\pi}_t + \tau_g \hat{y}_t^g$ (with $\tau = 1.5$, $\tau_g = 0.5$), a policy that stabilizes inflation, and the optimal monetary policy under commitment.

Inflation behavior under optimal policy is essentially zero inflation. This occurs despite an output gap opening up for the first four quarters. This illustrates how agency costs break the divine coincidence present in the standard DNK model according to which both inflation and output gap can be stabilized simultaneously with technology shocks. Again this is because technology shocks lead to movements in the credit market distortion that are akin to endogenous markup shocks. While our model embeds a policy trade-off even for technology shocks, this trade-off is largely resolved in favor of stabilizing inflation (as suggested by the extremely large inflation weight in the model's welfare criterion). In contrast, under the Taylor rule the inflation rate moves sharply upward (after the initial period), thus mitigating the increase in the gap. Under all three policies, the technology shock decreases the credit distortion and

9. The latter effect is a consequence of the sticky price assumption. In a sticky wage model the monetary expansion would cause profits and share prices to rise. This suggests that the response of the credit distortion to a monetary shock depends upon the nature of the monetary transmission mechanism, for example, sticky prices versus sticky wages.

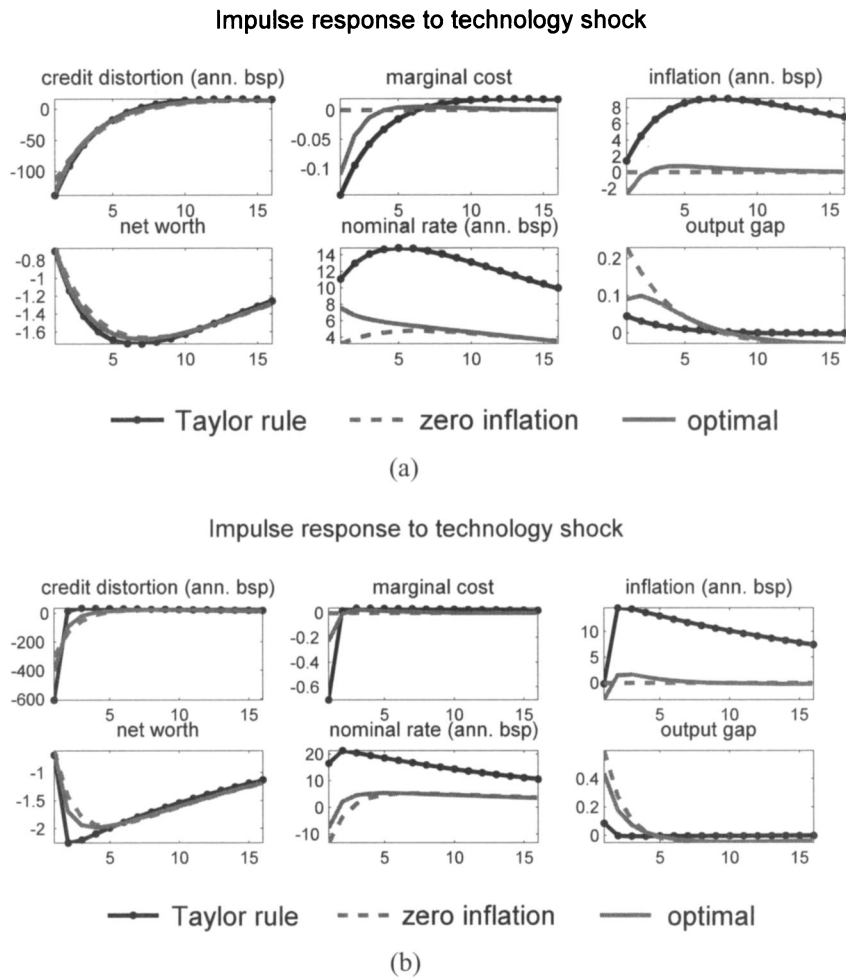


FIG. 2. (a) Impulse Response to a Technology Shock for $b = 0.2$. (b) Impulse Response to a Technology Shock for $b = 0.67$.

NOTE: See Figure 1.

net worth declines and responds in a hump-shaped manner. Evidently, the negative technology shock drives down the demand for credit more sharply than the decline in net worth. Although we do not report it for space considerations, the opposite result occurs with $\sigma > 1$.

For the alternative calibration of $b = 2/3$ in Figure 2(b), the negative technology shock has a much bigger impact on the credit market distortion than it does for our baseline $b = 0.20$ calibration. But the central message remains: inflation targeting is nearly optimal.

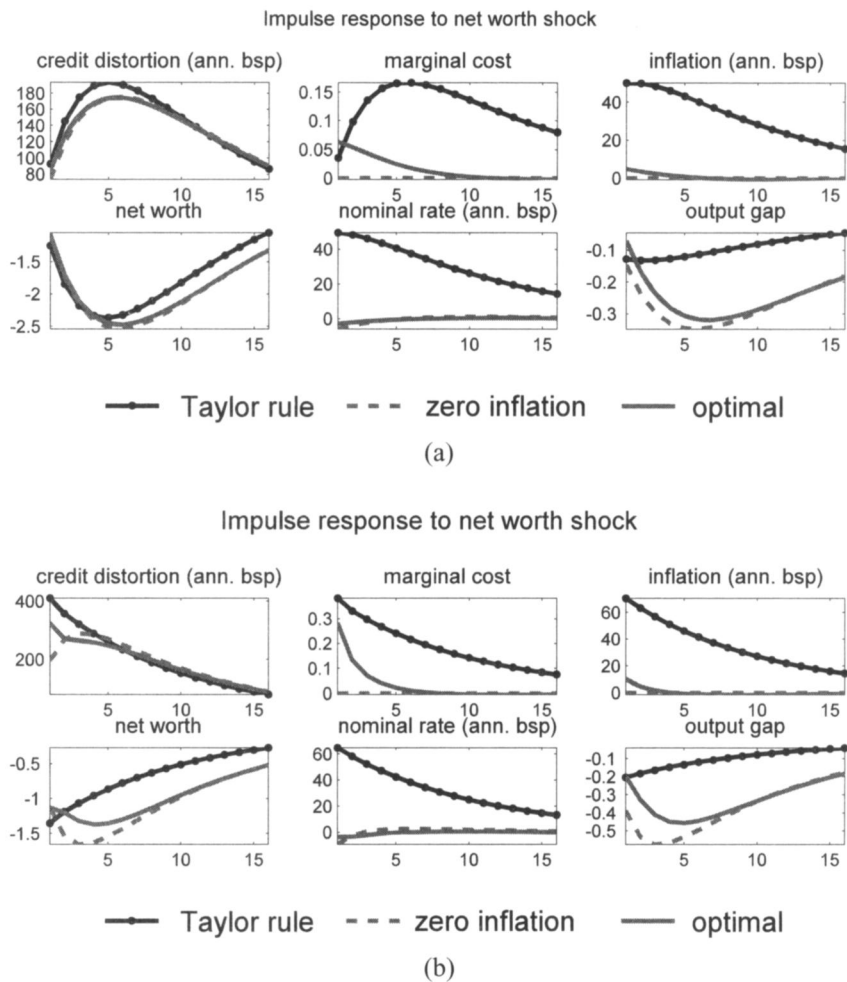


FIG. 3. (a) Impulse Response to a Net Worth Shock for $b = 0.20$. (b) Impulse Response to a Net Worth Shock for $b = 0.67$.

NOTE: See Figure 1.

Figures 3(a)–(b) present the complementary impulse responses to an exogenous negative net worth shock of 1%. The immediate impact of the shock is to sharply increase the credit distortion. This increase leads to a decrease in output and the gap. Under the Taylor rule the central bank allows for a larger inflation movement (compared to optimal policy) that mitigates the decline in the gap. In sharp contrast, the zero inflation policy leads to (of course) no inflation movement but a much larger decline in the gap. Optimal policy is somewhere in between, with more modest movements in inflation and the gap. Stabilizing inflation is not optimal because net worth shocks manifest themselves and endogenous markup shocks. Curiously,

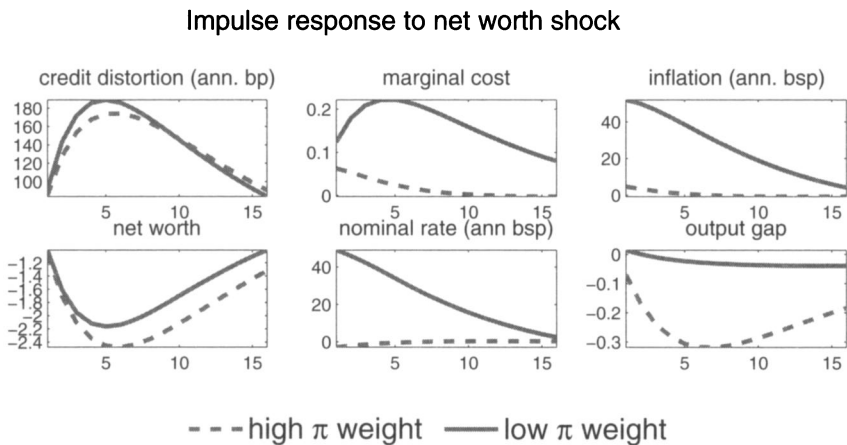


FIG. 4. Impulse Response to a Net Worth Shock with Alternative π -Weights ($b = 0.2$).

NOTE: See Figure 1.

optimal policy results in the impact effect on the gap to be nearly identical to the Taylor rule, but the later dynamics are similar to the zero-inflation policy. This similarity arises from the similarity in the net worth behavior under optimal and zero-inflation policy. Finally, note that because of strategic complementarities, our calibration of $\sigma = 0.16$ and $\theta = 0.47$ implies that the Phillips curve trade-off between inflation and the output gap is very steep. Even a small increase in inflation implies that optimal policy can close about two-thirds of the output gap that opens up in the first few quarters under a policy of full inflation stabilization.

The larger b calibration in Figure 3(B) leads to a sharper increase in the credit distortion and thus a larger effect on the gap and inflation. The subsequent dynamics are also altered as much of the hump-shaped behavior is eliminated, especially under the Taylor rule policy.

As noted earlier, optimal policy implies very small inflation movements because of the large weight on inflation in the welfare criterion. This suggests a natural sensitivity analysis in which we lower the weight on inflation. The weight on inflation in the criterion is given by $\frac{(\varepsilon-1)}{\lambda} = \varphi$. We hold λ in the Phillips curve constant but lower φ (and hence ε) so that the weight on inflation is equal to the weight on the output gap. We call this the “low π -weight criterion” in contrast to the benchmark “high π -weight criterion” in which the weight on inflation is nearly 300 times larger than the weight on the gap. Figure 4 presents the response to a negative net worth shock for the two optimal policies for the baseline case, $b = 0.2$. Interestingly, the behavior of the credit distortion is similar under the two criteria. From (33), the central bank can accommodate this movement in the risk premium either by increasing inflation or lowering the gap. Since λ is relatively small, fairly modest movements in inflation are enough to satisfy (33). Alternatively, stable inflation is possible only with movements in the gap that are comparable to the movement in the risk premium (since $\alpha/(\sigma + \theta)$

TABLE 1
WELFARE LOSSES UNDER DIFFERENT MONETARY POLICIES

	Technology shocks		Net worth shocks		Markup shocks	
	<i>b</i> = 0.2	<i>b</i> = 0.67	<i>b</i> = 0.2	<i>b</i> = 0.67	<i>b</i> = 0.2	<i>b</i> = 0.67
Optimal commitment	$4\sigma_a^2$	$19\sigma_a^2$	$59\sigma_n^2$	$109\sigma_n^2$	$284\sigma_\pi^2$	$298\sigma_\pi^2$
	$2\sigma_a^2$	$10\sigma_a^2$	$25\sigma_n^2$	$47\sigma_n^2$	$303\sigma_\pi^2$	$311\sigma_\pi^2$
	$0\sigma_a^2$	$0\sigma_a^2$	$1\sigma_n^2$	$2\sigma_n^2$	$321\sigma_\pi^2$	$321\sigma_\pi^2$
Inflation stabilization	$5\sigma_a^2$	$22\sigma_a^2$	$63\sigma_n^2$	$123\sigma_n^2$	$346\sigma_\pi^2$	$356\sigma_\pi^2$
	$2\sigma_a^2$	$11\sigma_a^2$	$25\sigma_n^2$	$50\sigma_n^2$	$380\sigma_\pi^2$	$386\sigma_\pi^2$
	$0\sigma_a^2$	$0\sigma_a^2$	$1\sigma_n^2$	$2\sigma_n^2$	$416\sigma_\pi^2$	$416\sigma_\pi^2$
Taylor rule with gap	$78\sigma_a^2$	$148\sigma_a^2$	$1150\sigma_n^2$	$1454\sigma_n^2$	$5806\sigma_\pi^2$	$5980\sigma_\pi^2$
	$39\sigma_a^2$	$65\sigma_a^2$	$294\sigma_n^2$	$385\sigma_n^2$	$5761\sigma_\pi^2$	$5840\sigma_\pi^2$
	$15\sigma_a^2$	$16\sigma_a^2$	$1\sigma_n^2$	$2\sigma_n^2$	$5722\sigma_\pi^2$	$5725\sigma_\pi^2$
Taylor rule without gap	$87\sigma_a^2$	$105\sigma_a^2$	$74\sigma_n^2$	$145\sigma_n^2$	$371\sigma_\pi^2$	$491\sigma_\pi^2$
	$92\sigma_a^2$	$100\sigma_a^2$	$28\sigma_n^2$	$54\sigma_n^2$	$396\sigma_\pi^2$	$450\sigma_\pi^2$
	$98\sigma_a^2$	$98\sigma_a^2$	$1\sigma_n^2$	$2\sigma_n^2$	$420\sigma_\pi^2$	$422\sigma_\pi^2$

NOTES: The three entries in each cell correspond to α s = 0.5, 0.25, and 0.01. We assume that the TFP follows an AR(1) process with autoregressive coefficient 0.95, whereas net worth and markup processes are assumed to slightly less persistent with AR coefficients 0.9. Welfare losses are scaled by the variance of the innovation in the exogenous processes for TFP (σ_a^2), net worth (σ_n^2), and markup (σ_π^2), respectively. Each table entry is multiplied by 100.

is about one). Hence, if there are equal weights on inflation and the gap, the central bank will choose to inflate and avoid sharp declines in the gap. But if the weight on inflation is high enough (as it is in the benchmark calibration), then the central bank will curtail inflation movements and allow the gap to absorb the shock.

2.3 Quantifying Welfare Costs of Alternative Policies

In Table 1 we quantify the welfare losses for four different specifications of policy. Note that these losses are scaled by the variance of the innovations in the exogenous stochastic processes and are thus very small for typical calibrations of these variances, for example, $\sigma_a^2 = (0.007)^2$.

Along with the zero-inflation and optimal policy under commitment, we calculate the welfare losses for alternative interest rate rules. We set $\tau = 1.5$ and $\tau_g = 0.5$ (“with gap”) or $\tau_g = 0$ (“without gap”). We do not take a stand on the relative importance of the structural shocks, so we compute the welfare loss based on unconditional variances separately for each shock. Again, several comments are in order.

First, the agency cost distortions imply that the central bank cannot eliminate all welfare losses arising from technology shocks. Further, the presence of agency costs magnifies the welfare costs of all shocks, regardless of monetary policy. Second, as already suggested, inflation stabilization comes quite close to achieving the welfare level of the optimal monetary policy for all three shocks.

Third, the two Taylor rules do significantly worse than the zero-inflation policy. As noted earlier, the welfare function places great weight on inflation, and under either Taylor rule the volatility of inflation is sharply increased.

Fourth, and most interestingly, whether including the gap in the Taylor rule is efficacious depends upon the nature of the shock and the degree of agency costs. If agency costs are relatively unimportant (the bottom row in each entry), then responding to the gap is welfare improving if there are technology shocks because an increase in the gap response is equivalent to an increase in the inflation coefficient. However, this is not the case when there are markup shocks. When there are markup shocks, responding to the gap causes the volatility of inflation to increase relative to a rule that does not respond to the gap.

But agency costs add two interesting wrinkles to this result. First, agency costs manifest themselves as endogenous markup shocks when the Phillips curve is expressed in terms of the gap (see (33)). Hence, even if there are only technology shocks, the presence of agency costs can make responding to the gap a bad idea (see Table 1). Second, for the case of net worth shocks, responding to the gap is always a bad idea, and the welfare cost can be quite large depending upon the variance of net worth shocks and how important agency costs are. Thus, the endogenous markup shocks coming from agency costs can be quite important.

2.4 Optimal Policy in a Credit Crunch

U.S. dollar investment-grade corporate bond spreads have risen sharply over the course of the recent credit crunch. Spreads in sterling and euro-denominated bonds have behaved similarly. What does the model predict for output, inflation, and monetary policy when hit by shocks that produce risk premia movements of this magnitude? Since the model contains no direct implications for the risk premium, we assume that the model's underlying risk premium is given by the risk premium in the isomorphic CSV model. Using our baseline calibration of $\sigma = 0.16$ and $\theta = 0.47$, $\alpha = 0.5$, and $b = 0.2$ this implies that the risk premium is given by $0.43 * b * \phi_t$ (see the Appendix for details).

To answer such a question, we need to take a stand on the nature of the exogenous shocks and the stance of monetary policy. We assume that monetary policy is conducted optimally as developed earlier. To generate the risk premia movements, we pick a series of net worth innovations that induce the actual time series behavior of the BB+ risk premium from 1997:1 to 2009:3. As in our baseline calibration we assume that this series has a steady state of 330 bps. The autocorrelation in the net worth shocks is set to $\rho_n = 0.9$. After 2009:3 we assume that all innovations are zero and net worth evolves according to its $\rho_n = 0.9$ autocorrelation.

We compute the model's predicted behavior of the output gap, inflation, the risk premium, and the *ex ante* real rate under the optimal monetary policy, the policy that stabilizes inflation, and a Taylor rule with gap ($\tau = 1.5$ and $\tau_g = 0.5$). Movements in the risk premium manifest themselves as endogenous markup shocks in (33). Note that in the standard DNK model ($\alpha = 0$), movements in the risk premium have

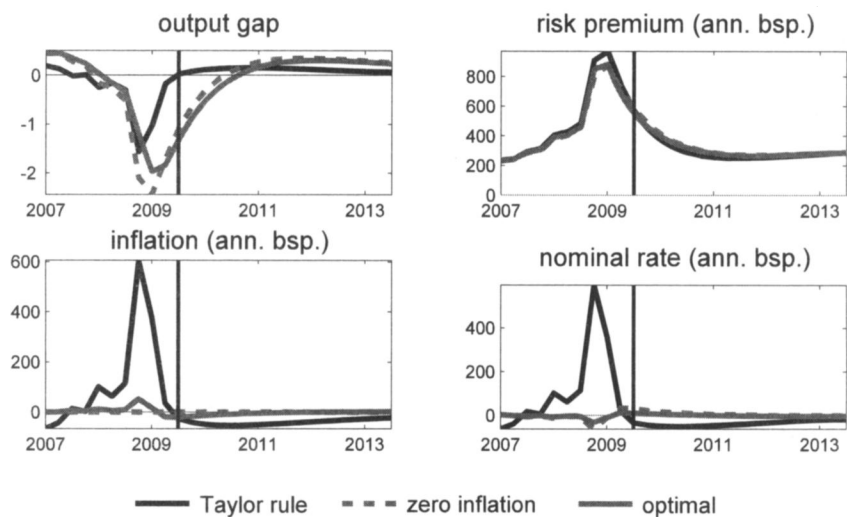


FIG. 5. Evolution with Net Worth Shocks.

NOTES: Evolution of the economy with net worth shocks only. Assuming optimal policy, net worth innovations chosen until 2009Q3 to match the actual behavior of the risk premium, zero innovations thereafter.

no effect. This exercise thus sheds light on the policy trade-offs embedded in the model.

Figure 5 shows the model behavior under optimal commitment and under full inflation stabilization. Under zero inflation, the gap and the risk premia are linked tightly by (33)

$$\hat{y}_t^g = -\frac{\alpha}{(\sigma + \theta)} b \hat{\phi}_t = -1.85 r p_t,$$

where the calculation uses our baseline calibration with $\alpha = 0.5$ that the risk premium is given by $0.43 * b * \phi_t$. That is, a zero-inflation policy causes the net worth shocks to feed entirely into movements in the gap. This is clearly too much as the welfare function (30) implies that the weight on the gap $(\sigma + \theta)$ is slightly less than the weight on the risk premium $(\frac{\alpha(1-\alpha)}{0.43^2(1+\theta)})$. A policy of zero inflation allows the gap to move by too much relative to the risk premium. This inflation movement is exacerbated under a Taylor rule. In contrast, optimal policy shifts more of the net worth shock on to a movement in the risk premium, and away from the gap. Optimal policy also allows for some movement in inflation (52 bps) to ameliorate the needed movements in the gap and the risk premium.

Overall, the predicted fall in output from this extraordinary rise in spreads seems somewhat small relative to the data. While outright declines in output are similar to the output gap declines in the picture (2%), the output gap in the data is larger since potential output growth is around $2\frac{1}{2}\%$. Of course, this experiment assumes that the only shocks hitting the economy are net worth shocks. But even given this, the

mapping from spreads to output depends crucially on the labor supply elasticity and the intertemporal elasticity of substitution. Smaller values of σ or θ would increase the output declines in the model.

3. CONCLUSION

This paper has developed a tractable model of credit frictions in an otherwise standard DNK model. A key implication of our model is that agency costs manifest themselves as endogenous mark-up shocks. Consequently, there generally is a trade-off between stabilizing inflation, the output gap and the risk premium, regardless of the type of shock hitting the economy—that is, there is no divine coincidence. Hence, the model provides a rationale for the central bank to consider credit frictions when setting policy. In particular, a measure of credit market tightness which we interpret as a risk premium enters directly into the central bank welfare criterion derived here. However, the DNK nature of the model implies that fluctuations in inflation are much more costly in welfare terms than variability of the output gap or the risk premium. Consequently, stabilizing inflation is near optimal even if agency costs are quite severe.

This stylized model highlights the supply effects of agency costs. That is, the risk premium enters the model only via the Phillips curve. Hence, compared to zero inflation, optimal policy will magnify the response of the risk premium to a net worth shock. A more elaborate model would also include capital accumulation, which would add a demand-side element to the model via changed demand for investment goods (or consumer durables). But our conjecture is that a model with both supply and demand forces will still result in optimal policy magnifying the movement of the risk premium in response to net worth shocks. Curiously, this model that endogenously introduces risk premia into the central bank's welfare criterion will lead the central bank to magnify risk premia movements compared to its inflation-targeting counterpart.

APPENDIX

A.1 The Model in Log-Deviations

$$\sigma \hat{y}_t + \theta \hat{L}_t = \hat{w}_t \quad (\text{A1})$$

$$\sigma \hat{y}_t + \theta \hat{u}_t = \hat{r}_t \quad (\text{A2})$$

$$\sigma (E_t \hat{y}_{t+1} - \hat{y}_t) = \hat{R}_t - E_t \hat{\pi}_{t+1} \quad (\text{A3})$$

$$\hat{q}_t = \beta E_t \hat{q}_{t+1} + (1 - \beta) E_t \hat{d}_{t+1} - \sigma (E_t \hat{y}_{t+1} - \hat{y}_t) \quad (\text{A4})$$

$$\hat{e}_t + \hat{q}_t = \hat{z}_t + \hat{y}_t + \Lambda \hat{\phi}_t \quad (\text{A5})$$

$$\hat{w}_t + \hat{L}_t = b(\hat{e}_{t-1} + \beta q_t + (1 - \beta)\hat{d}_t + \hat{n}_t) + (1 - b)(\hat{z}_t + \hat{y}_t) \quad (\text{A6})$$

$$\hat{z}_t + \hat{y}_t = \hat{r}_t + \hat{u}_t \quad (\text{A7})$$

$$\hat{y}_t = \hat{a}_t + (1 - \alpha)\hat{u}_t + \alpha\hat{L}_t \quad (\text{A8})$$

$$\hat{\pi}_t = \lambda\hat{z}_t + \beta E_t \hat{\pi}_{t+1} + \lambda\epsilon_t^\pi \quad (\text{A9})$$

$$\hat{d}_t = \hat{y}_t - (\varepsilon - 1)\hat{z}_t \quad (\text{A10})$$

$$b\hat{\phi}_t = \hat{z}_t + \hat{y}_t - \hat{w}_t - \hat{L}_t, \quad (\text{A11})$$

where $\Lambda \equiv \frac{F'}{F} \approx (b - 1) \leq 0$, and we have used the fact that $\hat{p}_t = \hat{z}_t + \hat{a}_t$ and $\hat{c}_t = \hat{y}_t = \hat{a}_t + \hat{x}_t$. The system is closed with a policy rule for the nominal interest rate.

A.2 Steady-State and Subsidies

In the steady state we will choose subsidies for L-wages (w_{sub}) and u-wages (r_{sub}) to achieve the efficient steady state. In particular we have:

$$B_1 c_{ss}^\sigma L_{ss}^\theta = w_{ss} (1 + w_{\text{sub}})$$

$$B_2 c_{ss}^\sigma u_{ss}^\theta = r_{ss} (1 + r_{\text{sub}})$$

$$Q_{ss} = \frac{\beta}{1 - \beta} D_{ss}$$

$$z_{ss} = P_{ss} = \frac{\varepsilon - 1}{\varepsilon}$$

$$D_{ss} = X_{ss} (1 - z_{ss})$$

$$P_{ss} MPL_{ss} = w_{ss} (1 + b\phi_{ss})$$

$$P_{ss} MPU_{ss} = r_{ss}$$

$$e_{ss} Q_{ss} = \gamma \alpha P_{ss} x_{ss} \phi(\phi_{ss})$$

$$[e_{ss} (Q_{ss} + D_{ss})]^b (a P_{ss} x_{ss})^{1-b} = w_{ss} L_{ss}$$

$$c_{ss} = x_{ss}.$$

We pick the subsidies so that the first-best level of output is achieved

$$B_1 c_{ss}^\sigma L_{ss}^\theta = MPL_{ss} \frac{z_{ss}}{(1 + b\phi_{ss})} (1 + w_{sub}) = MPL_{ss}$$

$$B_2 c_{ss}^\sigma u_{ss}^\theta = MPU_{ss} z_{ss} (1 + r_{sub}) = MPU_{ss}.$$

A.3 Comparison to a CSV Model

The collateral constraint in the text (equation (6)) is quite similar to a more elaborate costly state verification (CSV) framework in which the CSV problem arises for the production of the output good (see Carlstrom and Fuerst 1998, 2001). Suppose that there are two sectors that produce input u_t and input L_t with a linear technology using the two labor inputs purchased from the household. These inputs are then sold on to the firms at the next stage that use a Cobb–Douglas aggregator (see (5)) to combine the two inputs. The product at this stage is then sold on to the sticky price firms at price p_t . The u -technology is not subject to informational frictions so that the wage here (r_t) will also be the price at which the input is sold on to the next stage. But the L -technology is subject to a CSV problem so that production is constrained by entrepreneurial net worth. The entrepreneur purchases labor input L_t and transforms this into intermediate input $\omega_t L_t$, where ω_t is an idiosyncratic, unit-mean, lognormal random variable that is costlessly observed by the entrepreneur but all others must pay a monitoring cost that is proportional to the project size. Let $\phi(\omega_t)$ and $\phi(\omega_t)$ denote the cdf and pdf of the idiosyncratic shock, respectively. The variable ω_t denotes the critical level of ω_t . For realizations below $\bar{\omega}_t$, the entrepreneur defaults, and vice versa. Because of the CSV informational friction, the entrepreneur's production will sell at a price equal to $s_t w_t$ where $s_t > 1$, is the markup over the labor cost. The share of the project outcome paid to the entrepreneurs is $f(\bar{\omega}_t)$ while $g(\bar{\omega}_t)$ denotes the share paid to the lender (ultimately the households). These are defined as:

$$f(\bar{\omega}) \equiv \int_{\bar{\omega}}^{\infty} \omega \phi(\omega) d\omega - [1 - \Phi(\bar{\omega})] \bar{\omega} \quad (\text{A12})$$

$$g(\bar{\omega}) \equiv \int_0^{\bar{\omega}} \omega \phi(\omega) d\omega + [1 - \Phi(\bar{\omega})] \bar{\omega} - \Phi(\bar{\omega}) \mu, \quad (\text{A13})$$

where μ is the monitoring cost. The optimal contract is then given by choosing L_t and $\bar{\omega}_t$ to solve the following:

$$\max s_t w_t L_t f(\bar{\omega}_t) \text{ subject to } s_t w_t L_t g(\bar{\omega}_t) \geq w_t L_t - n w_t. \quad (\text{A14})$$

The optimal contract is thus given by

$$s_t = \left[1 - \mu \Phi(\bar{\omega}_t) + \phi(\bar{\omega}_t) \frac{f(\bar{\omega}_t)}{f'(\bar{\omega}_t)} \right]^{-1}, \text{ or implicitly } \bar{\omega}_t = \Omega(s_t). \quad (\text{A15})$$

$$w_t L_t = n w_t \left[\frac{1}{1 - s_t g(\bar{\omega}_t)} \right] \equiv n w_t \mathcal{L}(s_t), \quad (\text{A16})$$

where the implicit function $\mathcal{L}(s_t)$ is the ratio of project size to net worth, “the leverage ratio.” The financial friction induces a labor market distortion as in the paper. In particular, the two labor choices are given by

$$\alpha p_t x_t = w_t L_t s_t \quad (\text{A17})$$

$$(1 - \alpha) p_t x_t = r_t u_t. \quad (\text{A18})$$

This CSV model bears close comparison to the model of the paper. Note first that in comparison to (7)–(8) in the text, we have that $s_t = (1 + b\phi_t)$. More to the point, (A16) is quite similar to the paper’s collateral constraint (6). Define the leverage ratio to be

$$\text{leverage} = \frac{w_t L_t}{n w_t} = \left[\frac{1}{1 - s_t g(\bar{\omega}_t)} \right] \equiv \mathcal{L}(S_t). \quad (\text{A19})$$

Let ξ denote the leverage elasticity: $\frac{d \ln(\mathcal{L})}{d \ln(s_t)} = \xi$. Hence, using (A17), we can write (A16) in log deviations

$$w_t + L_t = \frac{1}{1 + \xi} n w_t + \frac{\xi}{1 + \xi} (p_t + x_t). \quad (\text{A20})$$

In comparison to the text, this implies that $b = \frac{1}{1 + \xi}$.

The size of ξ depends upon the calibration of the CSV model. For $\mu = 0.15$, a steady-state bankruptcy rate of 1%, and a quarterly risk premium of 330/4 basis points, we have $\xi = 4$, which implies $b = 0.2$.

The risk premium in this CSV environment is given by

$$(1 + r p_t)(w_t L_t - n w_t) = \bar{\omega}_t s_t w_t L_t \quad (\text{A21})$$

or

$$(1 + r p_t) = \frac{\bar{\omega}_t}{g(\bar{\omega}_t)} = \frac{\Omega(s_t)}{g(\Omega(s_t))}. \quad (\text{A22})$$

As for the risk premium, from (A11) we can compute $\frac{drp}{d \ln(s_t)}$. Recall that the isomorphism implies $s_t = (1 + b\phi_t)$, so that $\frac{drp}{d \ln(s_t)} = \frac{drp}{db\phi_t}$. For the calibration outlined earlier we have $\frac{drp}{db\phi_t} = 0.43$. The steady-state level of s_t is 1.026 so that $b\phi_{ss} = 0.026$.

We close with two minor details. First, we have assumed that the steady state distortions are the same in the two models ($s_t = (1 + b\phi_t)$). This typically implies different values for the death rates in the two models. Second, the CSV model includes monitoring costs as part of the aggregate resource constraint. These are (of course) not part of the model of the text as bankruptcy does not occur in equilibrium. But these costs are calibrated to be quite small, so that this difference has no substantial effect on the dynamics of the two models.

A.4 Proofs

PROPOSITION 1. *The gross inflation rate that maximizes a present value welfare criterion is unity in the steady state regardless of whether subsidies exist that make the steady state efficient.*

PROOF. The policymaker maximizes $W_0 = E_t \sum_{t=0}^{\infty} \beta^t U_t$, where U is the period utility function, subject to the nonlinear equilibrium conditions. For this proof, we only need to consider those constraints that involve the inflation rate, that is, the resource constraint and the price setting equation. These are:

$$C_t = A_t L_t^\alpha U_t^{1-\alpha} \left(1 - \frac{\varphi}{2} (\pi_{t-1})^2 \right)$$

$$((\varepsilon - 1) - \varepsilon z_t + \varphi (\pi_t - 1) \pi_t) Y_t C_t^{-\sigma} = E_t \varphi \beta (\pi_{t+1} - 1) \pi_{t+1} Y_{t+1} C_{t+1}^{-\sigma}$$

Here, ε is the elasticity of the final goods bundler. Let λ_t^1 and λ_t^2 , respectively, denote the Lagrange multipliers on these constraints. The first-order condition of the Ramsey planner problem for choice of π_t evaluated at the steady state is

$$0 = \lambda_{ss}^1 \varphi (\pi_{ss} - 1) A_{ss} L_{ss}^\alpha U_{ss}^{1-\alpha}.$$

The resource constraint is a binding constraint on the policymaker, $\lambda_{ss}^1 > 0$; consequently, the optimal steady state gross inflation rate is unity. Since none of the constraints used in this proof depend on subsidies, the results holds regardless of whether subsidies exist that render the steady state efficient. ■

PROPOSITION 2. *A quadratic approximation to the welfare function around a steady state with zero inflation is given by*

$$\frac{U(t) - U^*(t)}{U_c(ss) c_{ss}} \approx -\frac{1}{2} \left[\frac{(\varepsilon - 1)}{\lambda} \hat{\pi}_t^2 + (\sigma + \theta) (\hat{y}_t^g)^2 + \frac{\alpha(1 - \alpha)}{1 + \theta} (b\hat{\phi}_t)^2 \right],$$

where $U^*(t)$ denotes welfare in a first best economy without agency or price adjustment costs. Using (19), we can alternatively express this in terms of the two

underlying distortions

$$\begin{aligned} \frac{U(t) - U^*(t)}{U_c(ss) c_{ss}} \approx & -\frac{1}{2} \left[\frac{(\varepsilon - 1)}{\lambda} \hat{\pi}_t^2 + \frac{a[\theta + \alpha + \sigma(1 - \alpha)]}{(1 + \theta)(\sigma + \theta)} (b\hat{\phi}_t)^2 \right. \\ & \left. + \frac{1}{(\sigma + \theta)} \hat{z}_t^2 - \frac{2\alpha}{(\sigma + \theta)} b\hat{\phi}_t \hat{z}_t \right]. \end{aligned}$$

PROOF. For ease of exposition, we derive the welfare function in terms of absolute deviations and then convert to percentage deviations in the last step. We define $\tilde{X}_t \equiv X_t - X_{ss}$ as absolute deviations from the steady state and $\hat{X}_t \equiv (X_t - X_{ss})/X_{ss}$ as percentage deviations. In a first step, we take a quadratic approximation to the utility functional

$$\begin{aligned} U(t) - U(ss) \approx & U_c \hat{c}_t + U_L \hat{L}_t + U_u \hat{u}_t + \frac{1}{2} (U_{cc} \hat{c}_t^2 + U_{LL} \hat{L}_t^2 + U_{uu} \hat{u}_t^2) \\ & + U_{cL} \hat{c}_t \hat{L}_t + U_{cu} \hat{c}_t \hat{u}_t + U_{Lu} \hat{L}_t \hat{u}_t. \end{aligned}$$

The resource constraint is given by

$$c_t = A_t L_t^\alpha u_t^{1-\alpha} \left[1 - \frac{\varphi(\pi_t - 1)^2}{2} \right] \equiv A_t f(L_t, u_t) \left[1 - \frac{\varphi(\pi_t - 1)^2}{2} \right].$$

A quadratic approximation to this expression is given by: (note that $\pi_{ss} = 1$)

$$\begin{aligned} \tilde{c}_t \approx & c_{ss} \tilde{A}_t + f_L(ss) \tilde{L}_t + f_u(ss) \tilde{u}_t + \frac{1}{2} (f_{LL}(ss) \tilde{L}_t^2 + f_{uu}(ss) \tilde{u}_t^2 - c_{ss} \varphi \tilde{\pi}_t^2) \\ & + f_{LA}(ss) \tilde{L}_t \tilde{A}_t + f_{uA}(ss) \tilde{u}_t \tilde{A}_t + f_{Lu}(ss) \tilde{L}_t \tilde{u}_t. \end{aligned}$$

Since we have assumed separable utility, and that the steady-state is efficient, we have

$$\begin{aligned} U(t) - U(ss) \approx & \frac{1}{2} (U_{cc} \tilde{c}_t^2 + U_{LL} \tilde{L}_t^2 + U_{uu} \tilde{u}_t^2) + \frac{1}{2} U_c (f_{LL}(ss) \tilde{L}_t^2 \\ & + f_{uu}(ss) \tilde{u}_t^2 - c_{ss} \varphi \tilde{\pi}_t^2 + f_{LA}(ss) \tilde{L}_t \tilde{A}_t \\ & + f_{uA}(ss) \tilde{u}_t \tilde{A}_t + f_{Lu}(ss) \tilde{L}_t \tilde{u}_t) + tip \end{aligned} \quad (A23)$$

Here, *tip* denotes terms independent of policy. Since all of these terms are of second order, we now only need consider linear terms in the subsequent expansions. The equilibrium choice of u is given by

$$U_u(t) = U_c(t) f_u(t) P_t.$$

Log-linearizing this expression and imposing efficiency in the steady state, we have

$$\hat{u}_t = \frac{\alpha(1-\sigma)}{\alpha(1-\sigma) + (\sigma + \theta)} \hat{L}_t + \frac{1}{\alpha(1-\sigma) + (\sigma + \theta)} \hat{z}_t + \frac{(1-\sigma)}{\alpha(1-\sigma) + (\sigma + \theta)} \hat{a}_t.$$

We will use this expression to eliminate u from the welfare function (A23). Labor choice can be characterized this way

$$U_u(t) f_L(t) = U_L(t) f_U(t) (1 + b\phi_t).$$

Expanding this up to the first order (and imposing efficiency in the steady state) we have:

$$b\hat{\phi}_t = (\theta + 1) \hat{u}_t - (\theta + 1) \hat{L}_t.$$

The efficient response to shocks is easily found by setting $\hat{z}_t = \hat{\phi}_t = 0$.

$$\hat{L}_t^e = \frac{(1-\sigma)}{(\sigma + \theta)} \hat{a}_t$$

$$\hat{u}_t^e = \left(\frac{1-\sigma}{\sigma + \theta} \right) \hat{a}_t.$$

We now define the labor gap to be the gap between actual labor and efficient labor. Substituting this into (A23), simplifying and subtracting the same approximations in an efficient economy with flexible prices and no credit frictions (which is independent of policy), we have

$$\frac{U(t) - U^*(t)}{U_c(ss) c_{ss}} \approx -\frac{1}{2} \left[\varphi \hat{\pi}_t^2 + \frac{\alpha(1+\theta)(\sigma + \theta)}{\theta + \alpha + \sigma(1-\alpha)} (\hat{L}_t^g)^2 + \frac{1-\alpha}{\theta + \alpha + \sigma(1-\alpha)} \hat{z}_t^2 \right].$$

Next, we use $\lambda = (\epsilon - 1)/\varphi$ to rewrite the weight on inflation in terms of the slope of the Phillips curve. Using (17) and (18) to eliminate $\hat{z}_t^2$ and $(\hat{L}_t^g)^2$, we arrive at

$$\frac{U(t) - U^*(t)}{U_c(ss) c_{ss}} \approx -\frac{1}{2} \left[\frac{(\epsilon - 1)}{\lambda} \hat{\pi}_t^2 + (\sigma + \theta) (\hat{y}_t^g)^2 + \frac{\alpha(1-\alpha)}{1+\theta} (b\hat{\phi}_t)^2 \right].$$

The expression in terms of marginal cost and the risk premium is found by replacing $(\hat{L}_t^g)^2$ in terms of marginal cost and the risk premium. ■

PROPOSITION 3. *The optimal inflation target under commitment from a timeless perspective is given by*

$$\begin{aligned}\hat{\pi}_t = & \left[\frac{\alpha b \varepsilon (1 - \beta) - (1 + \beta \Lambda)}{(\varepsilon - 1)(1 + \beta \Lambda)} \right] \Delta \hat{y}_t^g - D \left[\frac{\alpha b \varepsilon (1 - \beta)(1 - \alpha)}{(\varepsilon - 1)(1 + \theta)(1 + \beta \Lambda)} \right] b \Delta \hat{\phi}_t \\ & - \left[\frac{\varepsilon(1 - \beta)[\alpha b(\sigma - 1) + (\sigma + \theta)\Lambda] + (1 + \theta)(1 + \beta \Lambda)}{(\varepsilon - 1)(\sigma + \theta)(1 + \beta \Lambda)(1 + \Lambda)} \right] \hat{X}_t \\ & + \left[\frac{\varepsilon(1 - \beta)[\alpha b(\sigma - 1) - (\sigma + \theta)] + (1 + \theta)(1 + \beta \Lambda)}{\beta(\varepsilon - 1)(\sigma + \theta)(1 + \beta \Lambda)(1 + \Lambda)} \right] \hat{X}_{t-1}.\end{aligned}$$

The variable $\hat{X}_t$ is a lag-polynomial in $\Delta \hat{\phi}_t$ and $\Delta \hat{y}_t^g$ defined via the recursion:

$$\hat{X}_t = \beta A E_t \hat{X}_{t+1} + E_t b \Xi_{t+1}$$

$$\Xi_t \equiv \frac{\alpha \beta (\sigma + \theta) (1 + \Lambda) [(1 + \theta) \Delta \hat{y}_t^g - D(1 - \alpha) b \Delta \hat{\phi}_t]}{(1 + \theta) (\sigma + \theta + \alpha b (1 - \sigma))}.$$

Recall that $A \equiv \frac{\alpha b(1-\sigma)-\Lambda(\sigma+\theta)}{\alpha b(1-\sigma)+(\sigma+\theta)} < 1$, $\Lambda \equiv \frac{F'}{F} \approx (b - 1) \leq 0$.

PROOF.

$$\max_{\{\hat{\pi}_t, \hat{\phi}_t, \hat{y}_t^g, \hat{e}_t\}_{t=0}^{\infty}} -\frac{1}{2} E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(\varepsilon - 1)}{\lambda} \hat{\pi}_t^2 + (\sigma + \theta) (\hat{y}_t^g)^2 + D \frac{\alpha(1 - \alpha)}{1 + \theta} (b \hat{\phi}_t)^2 \right] \quad (\text{A24})$$

subject to

$$\hat{\pi}_t = \lambda (\sigma + \theta) \hat{y}_t^g + \alpha \lambda b \phi_t + \beta E_t \hat{\pi}_{t+1} + \lambda \epsilon_t^\pi, \quad (\text{A25})$$

$$\hat{e}_t = \frac{1}{\beta} \left\{ \hat{e}_{t-1} - \varepsilon (1 - \beta) (\sigma + \theta) \hat{y}_t^g + [1 + \beta \Lambda - \alpha \varepsilon (1 - \beta) b] \hat{\phi}_t + n_t \right\}, \quad (\text{A26})$$

$$\begin{aligned}\hat{\phi}_t = & \frac{(1 + \theta)}{(1 + \Lambda)} E_t \Delta \hat{y}_{t+1}^g - \frac{(1 - \alpha b)}{(1 + \Lambda)} E_t \Delta \hat{\phi}_{t+1} \\ & - \frac{(\sigma - 1)(1 + \theta)}{(\sigma + \theta)(1 + \Lambda)} E_t \Delta \hat{a}_{t+1} - \frac{1}{(1 + \Lambda)} E_t n_{t+1}.\end{aligned} \quad (\text{A27})$$

Let $\mu_t^1, \mu_t^2, \mu_t^3$ denote the Lagrange multipliers associated with (A27)–(A29). The first-order conditions for $t > 0$ are

$$0 = -\frac{(\varepsilon - 1)}{\lambda} \hat{\phi}_t - \mu_t^1 + \mu_{t-1}^1 \quad (\text{A28})$$

$$0 = -(\sigma + \theta) \hat{y}_t^g + \mu_t^1 \lambda (\sigma + \theta) - \mu_t^2 \frac{(1 - \beta) \varepsilon (\sigma + \theta)}{\beta} - \mu_t^3 \frac{(1 + \theta)}{(1 + \Lambda)} + \mu_{t-1}^3 \frac{(1 + \theta)}{\beta (1 + \Lambda)} \quad (\text{A29})$$

$$0 = -\frac{D\alpha(1 - \alpha)}{1 + \theta} b \hat{\phi}_t + \mu_t^1 \alpha \lambda + \mu_t^2 \frac{[1 + \beta \Lambda - \alpha \varepsilon (1 - \beta) b]}{b\beta} - \mu_t^3 \frac{\alpha b + \Lambda}{b(1 + \Lambda)} + \mu_{t-1}^3 \frac{(\alpha b - 1)}{\beta b (1 + \Lambda)} \quad (\text{A30})$$

$$0 = -\mu_t^2 + E_t \mu_{t+1}^2. \quad (\text{A31})$$

Solve (A29) and (A30) for μ_t^1 and μ_t^2 . We scroll these expressions forward and backward to eliminate all leads and lags of μ_t^1 and μ_t^2 from the system. Equations (A28) and (A31) can now be expressed in terms of $\Delta \mu_t^3$, $\Delta \mu_{t-1}^3$, $\Delta \mu_{t+1}^3$, $\Delta \hat{y}_t^g$, $\Delta \hat{y}_{t+1}^g$, $b \Delta \hat{\phi}_t$, and $b \Delta \hat{\phi}_{t+1}$. The term $\hat{X}_t$ is proportional to $\Delta \mu_t^3$. We can then express the targeting criterion as in the proposition. ■

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